

The Research Content

of Investment Strategies

May 12, 2026

Red text marks placeholders or tasks before circulation. Purple text marks temporal quantities, current sample facts, or claims likely to change as the measurement design updates.

Abstract

Academic finance and investable products often describe related economic ideas in different languages. Academic papers study mechanisms, strategies, risks, and empirical relationships. Product documents describe objectives, exposures, instruments, benchmarks, rules, and investor-facing risks. This note develops a measurement design for comparing the two. The product side is represented as a small design graph, and the academic side is represented through nearby research-graph neighborhoods. Add the final abstract result here once the bridge sample and bridge-type distribution are fixed. In the current pilot, the main finding is not that products and papers can always be linked. The useful object is the type of link. Some product families map to academic work at the strategy level, some through a mechanism or rationale, some only through a shared market or motif, and some apparent matches are false keyword matches.

1 Introduction

Investment products often look familiar to academic finance, but only after translation. A covered-call fund looks close to the buy-write strategies studied in the literature. A target-date fund looks close to lifecycle portfolio choice. A buffer ETF is less direct: the product describes reference assets, FLEX options, caps, buffers, and outcome periods, while the academic neighborhood is closer to portfolio insurance and option-based payoff protection. These examples share finance ideas, but the product and the paper rarely describe the idea in the same form.

This paper studies that translation. Academic finance describes mechanisms, strategies, risks, and empirical relationships. Product documents describe objectives, exposures, instruments, benchmarks, payoff rules, horizons, and investor-facing risks. The research question is whether those two descriptions can be compared in a disciplined way. When an investment strategy is sold as a product, how much of its design can be linked to academic finance, and at what level: the same named strategy, the same mechanism, a shared design motif, a broad concept, or no credible bridge?

The current evidence points to bridge type as the central measurement object. In a first-pass adjudication of 135 product-academic candidate rows, 45 are classified as same-strategy bridges, 32 as mechanism bridges, 31 as shared-motif bridges, 15 as concept-only matches, and 12 as rejected matches. These counts should be read as pilot evidence from the current review queue, not as final population shares.

The paper’s measurement object is a product-research bridge. On the product side, I represent a prospectus as a small design graph. The graph records what the product seeks, what it holds or references, which instruments it uses, which rule or horizon defines the payoff, which benchmark it tracks, and which risks it discloses. On the academic side, I use research-graph neighborhoods from FrontierGraph, where papers are represented through concepts and relationships extracted from academic text. A bridge links the product graph to the academic neighborhood and classifies the connection.

The first evidence is clearest when product families are separated by bridge type. Managed futures, multifactor smart beta, and ESG-style products look closest to same-strategy bridges: the academic and product languages are relatively close. Inflation hedging, real assets, and volatility risk premium products look more like mechanism bridges: the product design uses a rationale or market mechanism studied in

the literature, even if the wrapper differs. Crypto products currently look more like shared market or shared motif bridges. Quality and low-volatility are useful boundary cases because the same words often retrieve adjacent academic meanings rather than the product’s intended strategy.

The question matters for several audiences. For academics, the bridge asks which research objects travel into market practice and which product objects remain understudied. For product designers, it asks whether a strategy is grounded in a studied mechanism or mainly packaged around a theme. For allocators, it separates products that implement a known strategy from products that share only surface vocabulary with the literature. For financial-innovation researchers, it creates a way to ask whether products follow academic formalization, precede it, or evolve in a parallel language.

The contribution is to define and measure a bridge between academic research objects and product design objects. Existing work on asset pricing, factor investing, funds, and financial innovation gives many reasons to care about how ideas become investable. The missing object is the translation layer itself. A keyword match cannot distinguish a product that discloses volatility risk from one that follows a volatility-control rule. A product graph can make that distinction visible and comparable to academic research graphs.

[**Placeholder** – add literature citations and exact positioning. Likely anchors: asset-pricing anomalies and factor investing; mutual funds and ETFs; financial innovation and product design; text-as-data and knowledge graphs in economics and finance; FrontierGraph as the academic graph source.]

[**Placeholder** – replace the provisional first-pass bridge counts with final reviewed counts once source-text adjudication is complete.]

[**Placeholder** – add final roadmap paragraph once the empirical design is chosen. The current preferred path is cross-sectional bridge measurement first, with product-entry timing as a later empirical layer.]

2 The Research Question

The broad question is:

How do ideas in academic finance appear in investable products, and

where do products use ideas that academic finance has not yet described in the same form?

There are several possible versions of this question.

1. A product may directly implement an academic strategy.
2. A product may use a mechanism that academic finance studies, even if the product wrapper is newer or more specific.
3. A product may share only a broad motif with academic work, such as rule-based exposure transformation or state-dependent allocation.
4. A product may mainly share words with the literature, without a substantive bridge.
5. A product may be industry-led, with little nearby academic treatment at the time it appears.

[**Placeholder** – add 2–3 concrete examples for each bridge type once the family-level review is stable.]

3 Data Objects

The product side comes from regulatory filings for U.S. mutual funds and exchange-traded funds. These filings include the text that a fund uses to describe itself to investors. The most useful fields for this project are the investment objective, principal investment strategy, benchmark or index description, and principal risks. These fields are useful because they usually say what the fund is trying to deliver, which assets or instruments it uses, which index or reference asset matters, and what risks the investor is asked to bear.

The unit on the product side is a product document. A document is tied to a fund or fund series and a filing date. It may describe a broad index fund, a target-date fund, a covered-call ETF, a buffer ETF, a crypto-linked product, or another strategy. I treat the document as a source of product design information, not as marketing copy to be scored as good or bad.

The academic side comes from FrontierGraph. FrontierGraph is a graph representation of academic research. Instead of treating a paper only as a title, abstract,

citation count, or journal publication, it represents the paper as a local graph. The nodes are research concepts, such as volatility, covered calls, target-date funds, liquidity, portfolio insurance, or bitcoin futures. The edges are relationships among those concepts, such as one object predicting another, one mechanism explaining another, or one variable affecting another. This representation makes academic papers comparable at the level of ideas and relationships.

The comparison uses both objects. A product document is turned into a product design graph. An academic paper, or a nearby set of academic papers, is represented through a research graph neighborhood. The bridge asks whether the product design and the academic neighborhood describe the same strategy, the same mechanism, a looser motif, only a shared concept, or no credible connection.

Add final sample facts here once the product-family screen, graph-extracted sample, and reviewed bridge sample are fixed.

[**Placeholder** – insert a table showing the current product sample construction: source filings, candidate screens, extracted rows, reviewed rows, and usable bridge examples.]

[**Placeholder** – add the exact FrontierGraph citation and describe the current academic source cut: field coverage, paper universe, graph extraction status, and how finance-relevant neighborhoods are selected.]

[**Placeholder** – define the product launch or first-observed date field if the paper studies timing between research and products.]

[**Placeholder** – document how SEC filing fields are mapped into objective, strategy, benchmark, and risk text, including any exclusions or cleaning rules.]

4 Representing Products As Design Graphs

A product graph is a compact description of what a fund says it does. The graph does not try to summarize the whole prospectus. It extracts the parts that describe the product’s design: the objective, the strategy, the instruments, the exposures, the rules, the benchmark, and the risks.

The reason to use a graph is that the same word can play different roles. A prospectus may mention options because the product uses options to create a payoff. It may mention options only as a risk. It may mention volatility because the strategy changes when volatility is high, because the product targets low-volatility stocks, or

because volatility appears in a generic risk disclosure. A keyword screen treats these cases as similar. A product graph separates them.

The current product graph uses nodes such as:

- product
- objective
- strategy
- instrument
- reference asset
- benchmark or index
- rule or horizon
- factor or style
- risk

The edges say how those objects relate to the product. The current edge types include:

- product seeks objective
- product uses instrument
- product references asset
- product tracks benchmark
- product has rule or horizon
- strategy targets factor or style
- product discloses risk

For example, a covered-call product might have an income objective, an equity or bitcoin reference exposure, and an option-writing strategy. A target-date fund might have a retirement date, underlying funds, and a glide path that changes the equity and fixed-income mix over time. A buffer ETF might have a reference asset, FLEX options, an outcome period, a cap, and a downside-protection rule. These are not just themes. They are product-design objects.

[**Placeholder** – add one worked product example as a small table or figure: product nodes, product edges, and source quotes.]

[**Placeholder** – add the final product-family screen rule once it is fixed. The rule should explain what evidence in the prospectus is sufficient for a product to enter a family.]

[**Placeholder** – add the final extraction scope: which product families are graph-extracted, why they are included, and what evidence is used from each filing.]

5 Bridge Types

A bridge is a claim that a product design object and an academic research object describe the same or related economic idea. The bridge can be close or loose. This distinction matters because product markets translate research ideas in several ways. Sometimes the product is almost the same object that academics study. Sometimes the product is a wrapper around a mechanism that academics study. Sometimes the connection is only a shared word.

A covered-call fund is a relatively close case. A covered-call strategy holds an asset, such as stocks or a stock index, and sells call options on that asset to earn option premium income.¹ Academic papers often study the same strategy under names such as covered calls or buy-write strategies. The product and the paper are therefore close at the strategy level.

A buffer or defined-outcome ETF is different. These products usually use options to create a payoff over a fixed period: the investor gets some upside up to a cap and receives some protection against losses up to a buffer or floor.² Academic papers may not study the exact ETF wrapper, but they do study related mechanisms: portfolio insurance, option replication, downside protection, and structured payoffs. The bridge is therefore less likely to be the same named strategy and more likely to be a mechanism or rationale bridge.

The current bridge vocabulary distinguishes these cases:

¹A call option gives the buyer the right to buy an asset at a specified price. A covered-call strategy sells call options while holding the underlying asset. The strategy usually gives up some upside in exchange for option premium income.

²A buffer ETF is designed to absorb a specified range of losses over an outcome period, often using option positions. A defined-outcome ETF states an intended payoff range, such as capped upside and partial downside protection, over a fixed horizon.

Table 1: Working bridge vocabulary

Bridge type	Meaning
Same strategy	The product and academic literature describe substantially the same operational strategy.
Same mechanism or rationale	The product uses an economic mechanism studied in the literature, but the product wrapper or implementation differs.
Shared motif	The product and literature share a broader structural pattern, such as rule-based exposure transformation, but not a close strategy match.
Concept only	The product and literature share a broad concept or keyword, but the connection is not yet substantively credible.
No bridge	The candidate match does not hold under review.

The categories are useful because they prevent a false sense of precision. If a product and a paper both mention volatility, the connection may be strong, weak, or accidental. A volatility-control product that adjusts exposure when volatility changes is closer to academic work on target-volatility rules. A product that merely lists volatility as a risk disclosure has a weaker connection. The bridge label records that difference.

The same logic applies to academic retrieval. A useful academic counterpart is not simply a paper with the same keyword. It should share a strategy, mechanism, rationale, or graph motif with the product. The bridge label is therefore part of the measurement, not an afterthought.

[**Placeholder** – decide whether “same mechanism” and “same rationale” should remain one category or be separated.]

[**Placeholder** – formalize academic counterpart retrieval from FrontierGraph. The method must state how candidate papers, concepts, and local graph neighborhoods are selected.]

[**Placeholder** – create a reproducible bridge-label adjudication table with product evidence, academic evidence, bridge type, and reason.]

6 Empirical Designs

The first empirical design is cross-sectional. For each product family, the paper asks what kind of bridge connects the product design to academic finance. Some families may have direct academic counterparts. Covered-call products, for example, can be compared to academic work on buy-write or covered-call strategies. Other families may connect through mechanisms rather than names. Buffer products can be compared to academic work on portfolio insurance, option replication, and downside protection, even if the exact ETF wrapper is not the object studied in the paper.

This design gives a simple first outcome: the bridge type. A product family can be close to academic finance at the strategy level, close at the mechanism level, related through a looser motif, connected only by a broad concept, or not credibly connected. The purpose is to compare product families on the type of academic connection they have, rather than to count whether a keyword appears in both places.

The second design is temporal. Once product entry dates are reliable, the same bridge object can be used to ask whether academic attention comes before, after, or alongside product entry. That design is attractive because it speaks directly to financial innovation: did academic research formalize the idea before the product appeared, or did product markets develop the object first? The timing design needs more data work before it can carry the paper.

[Placeholder – timing design parked for later: this requires reliable product first-observed dates, dated academic neighborhoods, and a rule for whether academic attention is measured by first paper, cumulative graph centrality, or local-neighborhood growth.]

The third design is interpretive. Even before timing is available, the product and academic graphs can show how the same broad finance idea is described differently. Products translate economic ideas into wrappers, payoff rules, benchmarks, income objectives, tax risks, and investor-facing exposures. Academic papers usually describe mechanisms, expected returns, risk channels, and empirical relationships. Comparing the two descriptions helps explain why a product may be research-linked even when it does not share the exact vocabulary of the paper.

The feasible first version is the cross-sectional and interpretive version. It compares product families by the kind of bridge they have to academic finance and uses product and academic graph language to explain why the bridge is close or loose. The

timing question can then be added once the product-entry data are strong enough.

[**Placeholder** – construct product-family first-observed dates from SEC series identifiers, fund names, tickers, and filing histories if the timing design enters the paper. Check whether first filing date is close enough to product launch date or whether we need an external launch-date source.]

[**Placeholder** – add the final cross-sectional bridge dataset definition: product family, product-design evidence, academic counterpart evidence, bridge type, and reason.]

7 Pilot Evidence

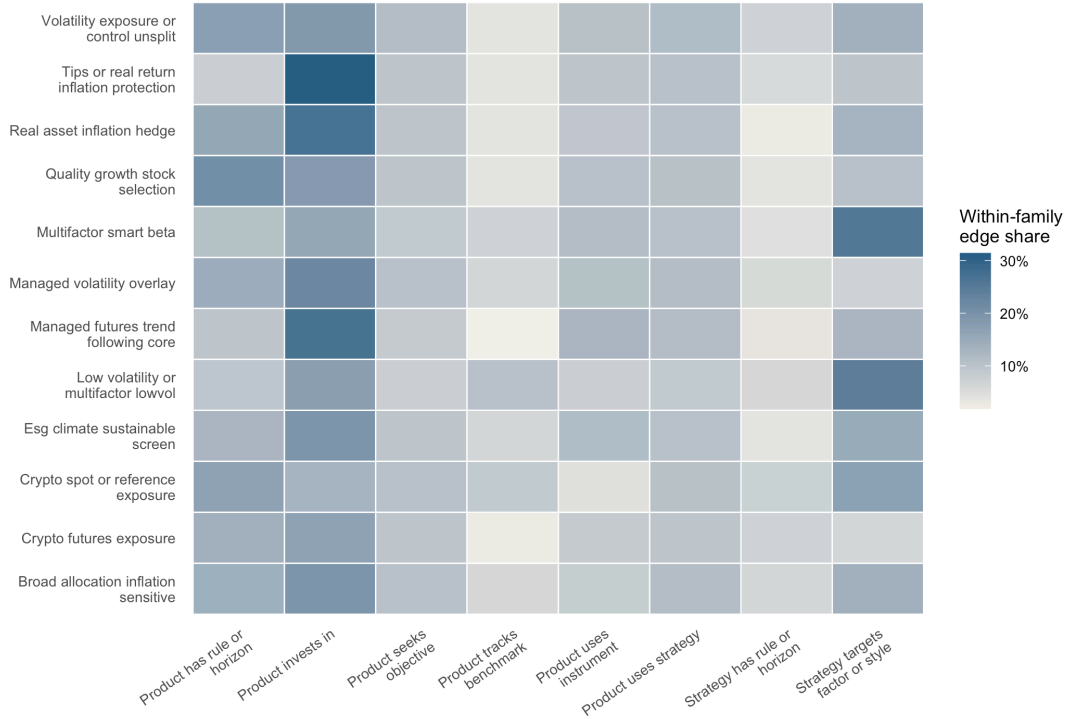
The current systematic product-graph tier contains 911 product documents across 12 product families. The extraction produced 9,987 product-side nodes, 9,655 product-side edges, and 7,331 bridge terms, with no parse failures. These quantities are current measurement facts, not final population estimates.

The first lesson is that prospectuses mostly describe product design, not academic mechanisms in miniature. Across the current product families, the recurring objects are holdings and exposures, instruments, objectives, rules and horizons, benchmarks, factor or style terms, and disclosed risks. This is useful. It means the product graph is capturing the language in which products are actually sold: what the fund holds, how it implements the strategy, what rule governs it, and what risk the investor is told to bear.

Figure 1 shows this point at the family level. The heat map reports the share of each family’s extracted edges that belong to the main product-graph relation types. Low-volatility and multifactor products are heavy in factor or style targeting. Crypto, TIPS, real-asset, and broad allocation products are more naturally described through what the product invests in. Managed-volatility and volatility-control products are more rule-like, with many edges connecting the product to a rule, horizon, instrument, or allocation mechanism.

The second lesson is that explicit state-conditioning is present but sparse. The product graph extracted 183 raw state-conditioning edges. A deterministic audit classifies 47 of those as plausible specific state edges, appearing in 29 of the 911 product documents. Most product documents therefore do not say, in a direct way, that the strategy changes under a named market state. They more often describe a design or exposure that may be linked to a state only after interpretation.

Figure 1: Product-graph signatures differ across product families

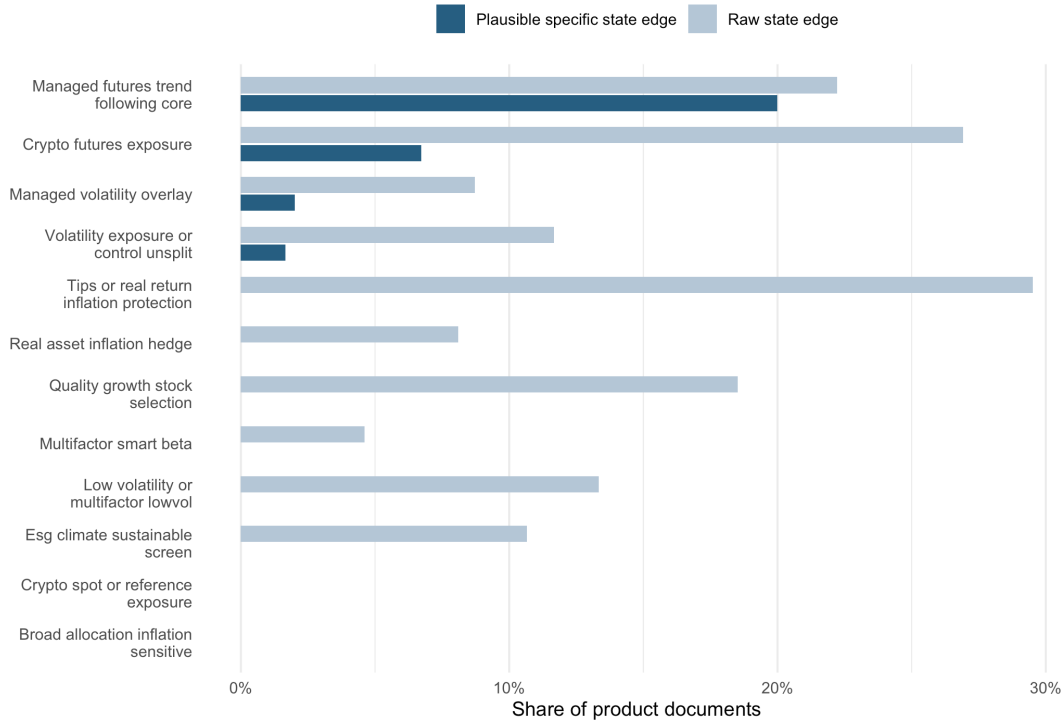


Notes: The figure uses the 911-document systematic product-graph tier. Each cell reports the within-family share of extracted product-graph edges in a given relation type. Darker cells indicate relation types that account for a larger share of the family graph. The figure is descriptive and uses the current extraction schema.

Figure 2 separates raw state-edge language from the narrower set of plausible specific state edges. Managed futures and trend-following products are the clearest current case: 18 of 90 documents contain plausible specific state-conditioning language. Crypto futures products also contain some specific state language, mainly around futures-market states such as contango and backwardation. Several other families have raw state-looking language, but little or no language that survives the stricter audit.

This result changes how the project should be framed. Conditioning remains useful, but it should not be the only bridge object. For many product families, the more natural bridge to academic finance is a strategy, instrument, payoff rule, benchmark, risk, or mechanism. The product-research bridge should therefore ask what kind of academic object is closest to the product graph, rather than asking only

Figure 2: Explicit state-conditioning is concentrated in a few product families



Notes: The figure uses the same 911-document systematic tier. A raw state edge is any extracted edge labelled as strategy-conditioned-on-state. A plausible specific state edge is a raw state edge that passes a deterministic audit for specific state language rather than generic market-condition language. The audit is used here as a measurement diagnostic, not as a final human-adjudicated label.

whether the product contains explicit regime-contingent language.

The first-pass bridge labels make this point directly. In the current 135-row review queue, 45 candidate rows are same-strategy bridges, 32 are mechanism bridges, 31 are shared-motif bridges, 15 are concept-only matches, and 12 are rejected matches. The label distribution is useful less as a population estimate than as a measurement result: product-academic overlap is heterogeneous.

The family-level pattern is also informative. Managed futures, multifactor/smart beta, and ESG/sustainable investing are the cleanest same-strategy cases in the current queue. Volatility-risk-premium, inflation-hedging, and real-asset products are closer to mechanism bridges. Crypto exposure usually shares a market or instrument with academic work, but not necessarily the same strategy. Quality and low-volatility show why the bridge cannot be a keyword match: the academic retrieval often finds

adjacent meanings such as earnings quality, fund-flow volatility, or minimum-variance mathematics.

Table 2: First-pass bridge labels differ by product family

Product family	Modal bridge	Strategy	Mechanism	Motif	Weak / rejected	Review rows	Reading
Managed futures / trend following	Same strategy	12	0	0	0	12	Product and academic languages meet at trend following and time-series momentum.
Multifactor smart beta	Same strategy	12	0	0	0	12	Product families map naturally to factor investing and smart-beta academic objects.
ESG / sustainable screen	Same strategy	12	0	0	0	12	The bridge is mostly a strategy or screen, though individual papers may differ in mechanism.
Volatility exposure / control	Mechanism	0	12	0	0	12	Academic links run through VIX, variance-risk premium, and volatility-risk-premium mechanisms.
Inflation-sensitive allocation	Mechanism	0	11	1	0	12	The bridge is mainly inflation hedging and inflation-risk exposure, not a single named product strategy.
Real-asset inflation hedge	Mechanism	0	9	3	0	12	Real assets connect to inflation-hedging mechanisms and asset-class motifs.
Crypto futures	Motif / market	0	0	12	0	12	Academic counterparts study bitcoin futures and futures-market structure more than the product strategy itself.
Crypto spot / reference	Motif / market	0	0	12	0	12	The bridge is mainly shared asset/market exposure.
TIPS / real return	Motif / market	0	0	3	0	3	Inflation-indexed bonds provide an instrument-level bridge; the current review set is small.
Quality / growth selection	Boundary	4	0	0	8	12	Retrieval often drifts to earnings quality or unrelated “quality” language.
Managed volatility overlay	Boundary	3	0	0	9	12	Some target-volatility links are useful, but generic risk-control retrieval is noisy.
Low-volatility / low-beta	Boundary	2	0	0	10	12	Academic retrieval often confuses low-volatility investing with flow volatility or minimum-variance mathematics.

Notes: Counts are from the current first-pass adjudication queue. “Weak / rejected” combines concept-only matches and rejected matches. The table is provisional: labels are useful for organizing the bridge typology, but selected cases still need source-text verification before final claims.

[**Placeholder** – quality check before stronger claims: verify 2–3 source papers per worked-example family and 2–3 product excerpts per worked-example family. The checks should confirm that the product graph, academic counterpart, and bridge label all survive source-text review.]

[**Placeholder** – review noisy families before circulation: quality / growth selection, low-volatility / low-beta, and managed-volatility overlay. These families are useful boundary cases, but they should not be described as final evidence without manual review.]

[**Placeholder** – classification decision: decide whether the current “shared motif” label should be split into “same instrument or market” and “broad motif.” Crypto

and TIPS are the main cases where this distinction may matter.]

7.1 Four Worked Examples

Managed futures is the cleanest same-strategy example. The product graphs describe futures-based strategies, rules, and trend-following language. The academic retrieval returns papers such as *Profitability of time series momentum*, *Time series momentum in the US stock market*, and *Optimal allocation of trend following strategies*. The bridge is close because the product and academic objects are both operational strategies: trend following and time-series momentum.

Multifactor smart beta gives a second same-strategy case. These product graphs are dominated by factor/style targeting, portfolio rules, and index-like implementation. The academic side retrieves factor investing and smart-beta papers. This is not merely a shared word. The product is sold as a rule-based factor implementation, and the academic neighborhood studies factor construction, return models, and factor-investing performance.

Volatility exposure is different. The best academic counterparts are not usually product-wrapper papers. They are mechanism papers around the VIX, variance risk premium, and volatility risk premium. This is a mechanism bridge: the product may use volatility-linked exposure or volatility-control rules, while the academic literature studies why volatility risk is priced and how volatility measures forecast returns or risk premia.

Quality and low-volatility show the boundary of the method. A keyword-only approach would treat many retrievals as academic support. The graph-and-adjudication view is stricter. “Earnings quality” papers are often about accounting or bank reporting, not quality investing. “Low volatility” papers may study capital-flow volatility or fund-flow volatility, not low-volatility equity products. These cases are useful because they show that product-academic comparison needs typed bridges, not vocabulary overlap.

[**Placeholder** – rerun bridge adjudication later with source-text or abstract evidence. The current bridge labels are first-pass labels based on titles, matched terms, retrieval tiers, product-family context, and deterministic risk flags.]

8 Interpretation

The bridge between research and products is not binary. Some product families look close to academic strategies. Others are commercial implementations of mechanisms studied in academia. Others share only broad motifs or markets. Some matches are only vocabulary matches and should be rejected. This is the central lesson from the current pass. The paper should therefore explain product–academic overlap as a typology of links, not as a yes-or-no match rate.

The current evidence supports a more precise statement: product graphs are useful because they separate product design objects before any comparison with academic research is attempted. A product can mention volatility as a risk, target low-volatility stocks as a style, follow a volatility-control rule, or use volatility-linked instruments. Those are different objects. A keyword screen tends to collapse them; a graph representation keeps them separate. The same logic applies on the academic side: a paper about earnings quality is not automatically a paper about quality investing, and a paper about capital-flow volatility is not automatically a paper about low-volatility equity products.

[**Placeholder** – add one paragraph explaining the professional interpretation: the method can help distinguish research-grounded strategy design from wrapper innovation or superficial theme language.]

[**Placeholder** – add one paragraph explaining the academic interpretation: the method can show which research objects travel into market practice and which product objects remain under-studied.]